



Islamic cryptocurrency integration for enhanced sustainable finance: evidence from time-frequency volatility transmission investigation

Emna Mnif, Yomna Daoud, Akrouf Zied & Anis Jarboui

To cite this article: Emna Mnif, Yomna Daoud, Akrouf Zied & Anis Jarboui (2024) Islamic cryptocurrency integration for enhanced sustainable finance: evidence from time-frequency volatility transmission investigation, *Journal of Chinese Economic and Business Studies*, 22:4, 435-453, DOI: [10.1080/14765284.2024.2388492](https://doi.org/10.1080/14765284.2024.2388492)

To link to this article: <https://doi.org/10.1080/14765284.2024.2388492>



Published online: 16 Aug 2024.



Submit your article to this journal [↗](#)



Article views: 111



View related articles [↗](#)



View Crossmark data [↗](#)

ARTICLE



Islamic cryptocurrency integration for enhanced sustainable finance: evidence from time-frequency volatility transmission investigation

Emna Mnif^a, Yomna Daoud ^a, Akrouit Zied^b and Anis Jarboui ^a

^aFaculty of Economic Sciences and Management of Sfax, Sfax University, Sfax, Tunisia; ^bEconomics department, College of Business, King Khalid University, Abha, Kingdom of Saudi Arabia

ABSTRACT

This study explores the potential of Islamic gold-backed cryptocurrencies in sustainable finance by focusing on volatility transmission between these cryptocurrencies and conventional digital assets like Bitcoin and PaxGold. Using a Time-Varying Parameter Vector Autoregression (TVP-VAR) approach, we analyze data from December 2019 to July 2023, including the COVID-19 pandemic period. Findings reveal a complex volatility network, with Bitcoin and HelloGold as major transmitters of spillover shocks, while X8X and PaxGold mainly serve as net recipients. Notable interactions between Bitcoin and Islamic gold-backed cryptocurrency markets show short-lived pairwise volatility interactions. Islamic gold-backed cryptocurrencies, net recipients of volatility, absorb market shocks, making them suitable for hedging against volatility. This can aid in diversifying portfolios, mitigating adverse market impacts, and informing risk management and regulatory approaches. Our research emphasizes the importance of Islamic finance in ethical and sustainable investment within the evolving cryptocurrency market.

ARTICLE HISTORY

Received 3 September 2023
Accepted 30 July 2024

KEYWORDS

Risk spillover; sustainability; Islamic gold-backed cryptocurrency; risk spillover

1. Introduction

The global landscape is grappling with escalating difficulties arising from heightened conflicts, the enduring impact of COVID-19, the threat of climate change, and ecological upheavals. The convergence of severe climate events and the ongoing repercussions of the pandemic has taken a profound toll on people's well-being (Jiang 2020). This confluence has, in turn, given rise to disruptions in social and economic spheres, amplified energy and food scarcities, and posed an ever-mounting menace to the environment. Additionally, the rapid advancement of technology and rampant speculation in financial markets have introduced new challenges. While technological innovations offer potential solutions to some of these issues, they also contribute to increased market volatility and raise concerns about data privacy and the digital divide. Market speculative behavior, fueled by easy access to information and high-speed trading technologies, exacerbates

financial instability and heightens the risks of asset bubbles and market crashes (García-Monleón, Danvila-Del-Valle, and Lara 2021).

With the imperative for sustainable economic and financial practices, Islamic finance emerges as a model of ethical and responsible financial conduct. With its foundations deeply rooted in Shariah law, Islamic finance principles inherently harmonize with the broader objectives of sustainability. This economic framework places significant emphasis on risk-sharing, ethical behavior, and prohibiting exploitative practices, including interest-based transactions. As we grapple with critical global issues such as climate change, social disparities (García-Monleón, Danvila-Del-Valle, and Lara 2021), and moral shortcomings within the financial sector, the fusion of Islamic finance and sustainability can spark profound and transformative change (Rabbani et al. 2021).

Islamic finance, grounded in equity, inclusivity, and ethical economic conduct principles, has garnered substantial recognition for its inherent compatibility with sustainable development objectives (Hassan et al. 2021). These principles have laid a sturdy foundation for exploring digital currencies that epitomize these values within the rapidly evolving landscape of technological advancement (Said, Annuar, and Hamdan 2019). Within this dynamic intersection, the ascendance of Islamic cryptocurrency technology presents an unprecedented opportunity to advance the tenets of the United Nations' Sustainable Development Goals (SDGs) and navigate the complex terrain of climate risk with newfound vigor and ingenuity (García-Monleón, Danvila-Del-Valle, and Lara 2021).

Islamic gold-backed cryptocurrencies have emerged as a novel and significant innovation in Islamic finance. Distinctively, their valuation is anchored to gold, paralleling certain mainstream digital currencies tied to physical assets like gold. Yet, these Islamic digital assets diverge fundamentally from their conventional peers. Their unique compliance with Shariah principles, which rigorously forbid speculative investments for Muslim investors, sets them distinctive. This adherence is achieved by pegging the cryptocurrency's value directly to a specified quantity of gold, thereby imbuing it with intrinsic value (Aloui, Hamida, and Yarovaya 2021). Islamic gold-backed cryptocurrencies exemplify this integration, joining the stability of gold with the ethical guidelines of Sharia law. This combination offers a compelling model for sustainable finance, aligning investment practices with the broader objectives of environmental stewardship, social equity, and economic resilience (Ali et al. 2022). Through their design, these assets offer a hedge against volatility and speculation, fostering a more stable investment environment conducive to sustainable development (Lahmiri and Bekiros 2019). The principles of Islamic finance prohibit interest (riba) and speculative behavior (gharar), aligning them with the sustainability goals that underpin both the SDGs and sustainable finance. Islamic gold-backed cryptocurrencies represent a convergence of these principles with modern financial instruments, offering a stable, non-speculative investment option that adheres to Shariah law. By linking these principles to the inherent stability provided by gold backing, Ali et al. (2022) considered that this form of cryptocurrency could significantly reduce volatility in financial markets by offering a safer, more sustainable investment compared to conventional cryptocurrencies.

Overall, Sustainable Development Goals (SDGs) aimed at achieving a better and more sustainable future for all. They address global challenges including poverty, inequality, climate change, environmental degradation, peace, and justice. In this analysis, the SDGs provide a broad context for understanding the potential impacts of economic policies and

practices, including those related to finance (Grover, Kar, and Ilavarasan 2019). Sustainable finance, on the other hand, refers specifically to the process of taking due account of environmental, social, and governance (ESG) considerations when making investment decisions in the financial sector; leading to increased long-term investments in sustainable economic activities and projects. In this study, sustainable finance is examined as a mechanism through which financial systems can contribute to the achievement of the SDGs by stabilizing economies and mitigating financial volatility. In this context, we explore how sustainable finance, influenced by the principles of the SDGs, can enhance economic stability by reducing the susceptibility to and impact of financial volatilities. In other words, this study aims to demonstrate how the integration of these concepts can offer a holistic approach to understanding and managing financial volatility in a way that supports sustainable development objectives.

For this purpose, this analysis delves into the dynamics of volatility transmission among Islamic and conventional cryptocurrencies, revealing a complex network mapping certain assets that act as transmitters or recipients of volatility. This behavior is critical for understanding the systemic implications of integrating such digital assets into the broader financial market, particularly in the context of sustainable finance. Maghyereh, Awartani, and Hassan (2018) suggest that adding gold to Islamic equity portfolios can optimize portfolio diversification. While some authors have expressed skepticism about the suitability of cryptocurrency as money from an Islamic perspective, others have highlighted its potential for use in Islamic finance (Shaikh et al. 2020; Siswantoro, Handika, and Mita 2020). Nevertheless, a few recent studies have investigated the role of non-Sharia-compliant cryptocurrencies (such as Bitcoin) in Islamic Sharia-compliant portfolios (Ali et al. 2022; Lahmiri and Bekiros 2019; Mnif, Mouakhar, and Jarboui 2022).

With the emergence of the COVID-19 pandemic, numerous studies have surfaced to investigate the impact of the global health crisis on risk contagion and return and volatility spillovers between Islamic and conventional cryptocurrency markets (Ali et al. 2022; Yousaf and Yarovaya 2022). One of the prevailing methods used is the DY connectedness approach, which utilizes VAR-based GFEVD proposed by Diebold and Yilmaz (2012, 2014) and its extended models in various domains. This study aims to examine the volatility spillover transmission among Islamic and conventional cryptocurrencies using a TVP-VAR frequency connectedness approach. The approach efficiently combines the essence of previous works by Baruník and Křehlík (2018) and (Antonakakis, Chatziantoniou, and Gabauer 2020). It allows for the decomposition of volatility connectedness into short-run and long-run components. It is achieved by simultaneously considering the time-varying coefficient and variance-covariance structure without losing observations or relying on arbitrary rolling windows. Our findings reveal a nuanced relationship between cryptocurrency volatility and sustainable finance, particularly highlighting how Islamic gold-backed cryptocurrencies can mitigate financial volatility while advancing sustainable finance principles. The study demonstrates that these cryptocurrencies exhibit lower volatility than traditional digital currencies and align with Sustainable Development Goals by promoting economic stability, ethical investing, and financial inclusion. This alignment suggests that integrating Islamic cryptocurrencies into broader financial practices can contribute significantly to achieving sustainability objectives, underscoring the potential of cryptocurrencies to serve as a bridge between innovative financial instruments and sustainable economic development. Considering the above, this study not only bridges the gap between Islamic finance and

the digital economy but also pioneers in elucidating the role of Islamic gold-backed cryptocurrencies within sustainable finance.

This paper offers several contributions and innovations. It contributes to the literature by examining the diversification benefits of Islamic digital assets in Islamic equity portfolios during increased uncertainty. Besides, by employing the cutting-edge TVP-VAR frequency connectedness method, our research offers unprecedented insights into the volatility transmission dynamics between Islamic and conventional digital assets during a period marked by significant global upheavals, including the COVID-19 pandemic. The distinctive findings highlighting Bitcoin and HelloGold as key transmitters of volatility, in contrast to X8X and PaxGold, which mainly absorb these shocks underscore the nuanced role these Islamic digital assets play in enhancing economic stability. This aligns with the Sustainable Development Goals and opens new avenues for ethical and sustainable investment strategies in the volatile cryptocurrency market. Thus, our contribution lies in advancing the academic discourse on Islamic digital assets and providing practical insights for investors and policymakers looking to navigate the complexities of sustainable finance in an increasingly digital world.

By harnessing the potential of Islamic cryptocurrencies, policymakers, financial institutions, and stakeholders stand to cultivate an ecosystem that aligns with Islamic values and contributes substantively to realizing the SDGs. As we navigate the dynamic landscape of finance and development, exploring Islamic cryptocurrency risk spillover as a tool for economic stability emerges as a compelling endeavor with far-reaching implications in portfolio management. Investors seek new investment opportunities with higher hedging effectiveness for their diversified portfolios due to the increased uncertainty caused by the COVID-19 pandemic.

The paper is structured as follows: [Section 2](#) provides a preliminary analysis of the data and the methodology. [Section 3](#) presents the empirical results, and [Section 4](#) summarizes the conclusions and implications of the study.

2. State of the art and hypothesis development

In an era marked by technological advancements and shifting paradigms in finance, a novel concept has emerged at the intersection of Islamic principles and digital innovation. These digital assets, designed to align with the ethical tenets of Islamic finance, hold the promise of not only revolutionizing the financial landscape but also serving as a catalyst for achieving the Sustainable Development Goals (SDGs). By integrating the principles of risk-sharing and ethical financial practices, Islamic cryptocurrencies have the potential to foster stable and resilient economies, thereby advancing the global agenda for sustainable development (Koeswandana and Sugino 2023). Islamic cryptocurrencies could mitigate the destabilizing impact of speculative bubbles and sudden market shocks. Several studies have focused on the response of cryptocurrency markets during crises and pandemics. The first strand of the literature has focused on market contagion and risk spillover to investigate the stability of financial markets. For instance, the analysis by Elsayed and Helmi (2021) reveals significant sensitivity to major political events like the Arab Spring and Gulf tensions. Concurrently, Lu et al. (2023) identified an increase in volatility correlations between the Indian equity market and

commodities post-COVID-19, suggesting commodities as strategic hedging tools during global crises. In the context of Islamic cryptocurrency markets, Mnif, Mouakhar, and Jarboui (2022) disseminated the lower-risk profile of Islamic cryptocurrencies during the COVID-19 pandemic, appealing to risk-averse, Shariah-compliant investors. Similarly, Yousaf and Yarovaya (2022) investigated the spillovers between Islamic gold-backed cryptocurrencies and Islamic equities during the pandemic, indicating the effectiveness of these assets in risk mitigation and portfolio diversification. Drawing from the insights into the behavior of gold-backed Islamic cryptocurrencies during market volatility and their potential as safe havens during the COVID-19 pandemic, alongside the broader implications of cryptocurrency adoption, the first hypothesis is formulated as follows:

H₁: The risk spillover between cryptocurrency markets is more pronounced during the pandemic.

The flight-to-safety behavior and increased allocations in gold and gold-backed cryptocurrencies have been observed in many past crises. The volatile and uncertain characteristics of cryptocurrencies have ignited a complex debate among Islamic scholars. The second strand evolves into discussions on the alignment of digital currencies with Islamic finance principles, examining whether their speculative aspects conform to or conflict with Islamic law's ethical and moral guidelines. Other scholars considered that the ethical underpinning of Islamic finance extends to the creation and utilization of these digital assets (Aloui, ben Hamida, and Yarovaya 2021; Siswantoro, Handika, and Mita 2020).

This scholarly inquiry holds significant practical implications, as it seeks to guide millions of Muslims in engaging with the contemporary financial world without compromising their religious beliefs. In this context, Hassan et al. (2021) argued that gold is highly productive as a haven against Islamic financial markets during the global financial crisis compared to the COVID-19 crisis. However, traditional Sharia-compliant assets (i.e. gold and Sukuk) have only been examined in the context of Sharia-compliant equity portfolios. A study by Abakah et al. (2023) demonstrated that Islamic stocks can hedge against Bitcoin volatility, while green bonds and clean energy indices offer stability against Fintech and AI stocks.

In addressing the intricate relationship between market volatility and sustainability, our literature review draws on recent empirical findings illuminating the role of innovative financial instruments in sustainable finance. Notably, the study of Ali et al. (2022) on gold-backed Islamic cryptocurrencies, such as OneGram Coin (OGC) and X8X Token (X8X), and their performance as safe havens during market distress, particularly during the COVID-19 pandemic, offers valuable insights into the stability these assets can bring to Islamic equity markets. Nevertheless, Yang et al. (2022) have analyzed sustainable practices across seven industries in G7 economies by exploring the Key determinants of ESG pillars such as green financing, clean energy, and green economic stability.

Therefore, we formulated the second hypothesis of this research as follows:

H₂: Islamic cryptocurrencies, as a novel Sharia-compliant asset class, enhance the diversification and stability of Islamic equity portfolios.

The spillover risk between conventional and Islamic cryptocurrencies have not been studied yet. Therefore, this paper aims to address this literature gap. Nevertheless, the cryptocurrency markets recently have gained considerable attention from retail and institutional investors, regulators, policy markets, Big Tech companies, and the media. The COVID-19 pandemic has further heightened interest in these markets. Recent research highlights the nuanced relationship between technological innovations in finance and sustainable investments. This research employs advanced risk assessment tools to underscore the potential of Islamic cryptocurrencies to mitigate losses in times of heightened market uncertainty. These findings are pivotal to our methodology as they exemplify the dual role of digital assets in enhancing financial stability and contributing to sustainable investment practices. By integrating such empirical evidence, our study aims to deepen the understanding of how volatility transmission mechanisms can be aligned with sustainability goals.

2.1. Data descriptions and methodology

In this study, we employ a Time-Varying Parameter Vector Autoregression (TVP-VAR) approach, which is particularly suited for examining the dynamic interplay between financial volatility and sustainable development goals over time. This method allows us to capture the evolving nature of financial markets and their responsiveness to shifts in sustainability-oriented practices. By incorporating macroeconomic variables and sustainability indicators, the TVP-VAR model adapts to changes in the economic environment, providing a nuanced understanding of how volatility transmission mechanisms operate under varying conditions of sustainable finance. The choice of TVP-VAR is justified by its ability to model the instability and changing relationships inherent in financial time series, which traditional static models may not capture effectively. This approach is particularly relevant to our study as it addresses the empirical gaps highlighted in previous research concerning the stability and predictiveness of financial relationships under the influence of sustainability practices. Through this methodology, we can rigorously test the hypothesis that sustainable finance practices contribute to economic stability, thereby enhancing the theoretical framework that connects economic volatility with sustainable development.

2.2. Data and variables presentation

This study meticulously examines a comprehensive daily dataset from 31 December 2019, to 31 July 2023, focusing on the volatile cryptocurrency market with a special lens on Islamic gold-backed cryptocurrencies such as HelloGold (HGT) and X8X Token (X8X), alongside conventional counterparts like PaxGold and the widely recognized Bitcoin (BTC). Sourced from Yahoo Finance, this dataset was chosen for its diversity and relevance, offering a unique opportunity to compare the stability and performance of Islamic cryptocurrencies against their conventional and digital peers within the same period. This selection underpins our objective to discern the nuances of volatility transmission across different market segments, providing a solid foundation for our analysis of the intersection of cryptocurrency volatility and sustainable finance principles.

2.3. Preliminary analysis

In this section, we interpret the summary statistics derived from the first log differences of the cryptocurrency price series while considering the ERS unit root test proposed by Elliott, Rothenberg and Stock (1996). These statistics offer insights into various aspects of the cryptocurrency series, including volatility, distribution characteristics, potential non-normalities, and the presence of unit roots. The mean return signifies the average change in value across each cryptocurrency's price series. As an illustration, BTC displays a mean return of -0.0004 , indicative of a slight average decline. This observation is complemented by a relatively elevated standard deviation of about 0.016 , underscoring BTC's inherent volatility in contrast to the other cryptocurrencies within our dataset. Skewness and kurtosis values delve into the shape and distribution characteristics of returns. A skewness of 1.595 for BTC suggests a positively skewed distribution, indicating that the cryptocurrency experienced more pronounced positive returns. A kurtosis value of 21.519 indicates that BTC's distribution possesses heavier tails than a normal distribution, implying heightened potential for extreme price movements. The Jarque and Bera (1980), with a significantly high value of $25,751.574$ for BTC, signifies a substantial deviation from a normal distribution (Table 1).

In the same way, the other cryptocurrency series display similar non-normality characteristics and potential outliers. The concept of time series stationarity drives the implementation of the TVP-VAR frequency connectedness approach in time series analysis. As a result, we embraced the ERS unit root test of Elliott, Rothenberg and Stock (1996), harnessing the capacity to detect serial correlation. The results of this test show that these variables are statistically stationary at a 1% significance level.

Figure 1 illustrates the variables' natural form, while Figure 2 plots the transformed series.

Figure 1 and Figure 2 show that the conventional cryptocurrency markets (HelloGold and X8X) experience the highest volatility throughout the data sample. Islamic cryptocurrency (X8X) encountered a prominent but not lasting volatility spike during the global pandemic. At the same time, the HelloGold market presents a relatively mild and notable decrease in volatility level with the outbreak of COVID-19.

2.4. TVP-VAR frequency

The TVP-VAR-based connectedness approach has been shown to effectively address some of the limitations of the rolling-window VAR methodology. This includes the problem of having to choose a rolling window size arbitrarily, the potential loss of observations, and the sensitivity of parameters to outliers. For these reasons, this paper employs a newly developed methodology known as the TVP-VAR frequency connectedness approach

Table 1. Descriptive statistics results for the daily series data.

	Mean	SD	Skewness	Kurtosis	Jarque-Bera	ERS
BTC	-0.0004	0.016	1.595^{***}	21.519^{***}	25751.574^{***}	-10.949^{***}
PaxGold	0.0002	0.0042	-0.190^{***}	7.313^{***}	2917.764^{***}	-12.365^{***}
HelloGold	-0.0005	0.0504	1.924^{***}	25.502^{***}	36196.104^{***}	-8.749^{***}
X8X	-0.0004	0.0522	-0.458^{***}	27.115^{***}	40055.106^{***}	-5.277^{***}



Figure 1. Cryptocurrency prices.

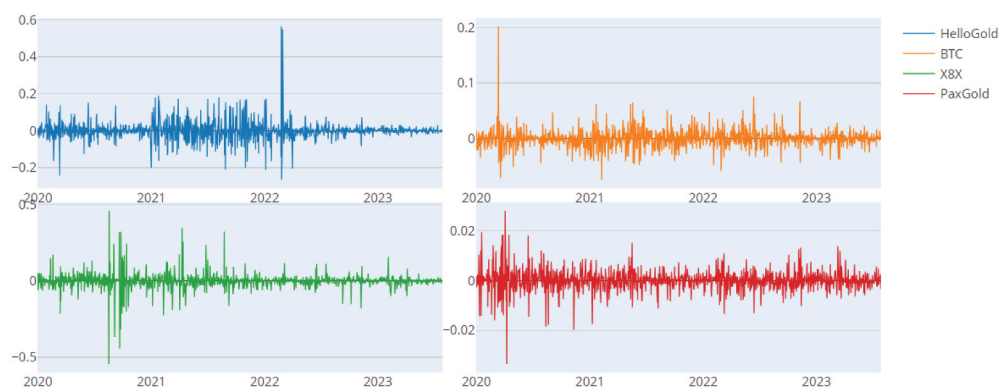


Figure 2. Cryptocurrency returns.

proposed by Chatziantoniou, Gabauer, and Gupta (2021). The outline of the TVP-VAR(p) can be described as follows:

$$x_t = \Phi_{1t}x_{t-1} + \Phi_{2t}x_{t-2} + \dots + \Phi_{pt}x_{t-p} + \varepsilon_t, \varepsilon_t \sim N(0, \Sigma_t) \quad (1)$$

The TVP-VAR(p) model is composed of y_t and ε_t , which are $N \times 1$ vectors, Σ_t , which is the $N \times N$ time-varying variance-covariance matrix, and Φ_{it} , $i = 1, \dots, p$, which represent the $N \times N$ time-varying VAR coefficient (Chatziantoniou, Gabauer, and Gupta 2021).

This formulation allows for a nuanced analysis of how shocks to one variable influence others over time, leveraging the generalized forecast error variance decomposition (GFEVD) to quantify connectedness measures. We extend this analysis into the frequency domain, integrating with the spectral representation to dissect the variance decompositions at various frequencies, thus offering insights into the cyclicity of interconnectedness. The frequency domain analysis utilizes the Fourier transform to compute the spectral density and the frequency specific GFEVD, enabling the examination of connectedness across different frequency bands.

By calculating net pairwise directional connectedness, total directional connectedness to and from others, and net total directional connectedness, along with the total averaged connectedness index, our methodology quantifies the dynamic interplay among

variables. It highlights the dominant channels of volatility transmission. This approach provides a comprehensive understanding of the systemic risk and market dynamics, essential for both researchers and practitioners concerned with financial stability and policy implications.

3. Empirical results and discussion

3.1. Total averaged volatility connectedness

Tables 2 , 3 and 4 present the total average connectedness among Islamic gold-backed and conventional cryptocurrency markets in various time horizons.

Table 2 displays the outcomes related to the overall mean interconnection across conventional and Islamic gold-backed cryptocurrency markets. In addition, Tables 3 and 4 delineate the short (1–5 trading days) and prolonged (5 days to infinity) elements.

Table 2 provides valuable insights into the relationships and interactions among the cryptocurrencies under study: HelloGold, BTC, X8X, and PaxGold. Notably, the diagonal values of the table represent the highest measure, as they reflect the relationship of each cryptocurrency with itself. The off-diagonal values present intriguing patterns. For instance,

HelloGold exhibits a low positive correlation with several other assets, indicating that its price movements are not strongly linked to those of the other assets. Specifically, HelloGold has a correlation of 16.06% with Bitcoin (BTC), which is considered low but still positive, suggesting that when BTC's price moves, HelloGold's price may move in the same direction, but the relationship is weak. With X8X, HelloGold's correlation is even lower at 0.41%, indicating an almost negligible relationship between their price movements. Similarly, HelloGold's correlation with PaxGold is also low at 0.50%, suggesting that the price movements of these two gold-backed assets are not closely related. These low correlations suggest that HelloGold behaves relatively independently compared to BTC, X8X, and PaxGold. Bitcoin (BTC) has a moderate positive correlation with HelloGold at

Table 2. Averaged volatility connectedness.

	HelloGold	BTC	X8X	PaxGold	FROM
HelloGold	83.02	16.06	0.41	0.50	16.98
BTC	16.06	81.83	0.94	1.17	18.17
X8X	5.12	12.71	81.02	1.15	18.98
PaxGold	1.34	5.01	0.74	92.91	7.09
TO	22.53	33.78	2.09	2.82	TACI
NET	5.55	15.60	−16.89	−4.26	15.30

Table 3. Averaged volatility connectedness in the short run (1 to 5 traded days).

	HelloGold	BTC	X8X	PaxGold	FROM
HelloGold	72.12	13.68	0.36	0.44	14.48
BTC	13.03	66.71	0.71	0.96	14.69
X8X	4.05	10.61	68.61	0.89	15.54
PaxGold	1.15	4.53	0.60	77.14	6.28
TO	18.23	28.81	1.67	2.28	TACI
NET	3.74	14.12	−13.87	−4.00	12.75

Table 4. Averaged volatility connectedness in the long run (5-infinite traded days).

	HelloGold	BTC	X8X	PaxGold	FROM
HelloGold	10.90	2.38	0.05	0.07	2.50
BTC	3.04	15.12	0.23	0.21	3.48
X8X	1.07	2.10	12.41	0.26	3.43
PaxGold	0.19	0.48	0.14	15.77	0.81
TO	4.30	4.96	0.42	0.54	10.22
NET	1.80	1.48	−3.02	−0.27	2.55

16.06%, indicating that there is some relationship between their price movements, but it is not strong. BTC's correlation with X8X is slightly lower at 12.71%, suggesting a weak relationship between their price movements. The correlation between BTC and PaxGold is even lower at 5.01%, indicating a very weak relationship. X8X has a weak correlation with both HelloGold and PaxGold, at 0.41% and 0.74% respectively, suggesting that its price movements are largely independent of these two assets. However, X8X has a slightly higher correlation with BTC at 12.71%, indicating a weak but somewhat more noticeable relationship. PaxGold, with its high self-correlation of 92.91%, shows very weak correlations with the other assets. Its correlation with HelloGold is 0.50%, with BTC is 5.01%, and with X8X is 0.74%, indicating that PaxGold's price movements are largely independent of those of the other assets. Overall, the correlations suggest that the price movements of these assets are generally weakly related to each other, with PaxGold showing the highest degree of independence.

The 'TO' row represents the Cumulative Total contribution for each cryptocurrency based on the variables measured in Table 2. The values in this row are calculated by summing up the values in each respective column for the corresponding cryptocurrency. The 'NET' row indicates net effects or changes derived from cryptocurrency interactions. These values reflect the net difference or impact after accounting for exchanges between the cryptocurrencies.

Initially, the composite gauge of total average connectedness, denoted as TACI, records a value of 15.3, indicating that, on average, around 15.30% of the anticipated error variance within the global financial markets network is attributable to the dissemination of shocks across these markets. The remaining 84.7% is accounted for by the unique aspects within each market. This level of risk transmission during COVID-19 is notably in line with several studies, such as Aloui, ben Hamida, and Yarovaya (2021) and Yousaf and Yarovaya (2022).

Moreover, the frequency decomposition analysis in Table 3 highlights that most Intermarket volatility interconnectedness, amounting to 12.75%, is attributed to short-term volatility transmission, while the long-term transmission documented in Table 4 accounts for a smaller share, at 2.55%.

Considering each asset's net total directional connectedness (NET) for short-term volatility (Table 3), Bitcoin exhibits the highest net transmission of volatility at 14.12%, followed by HelloGold at 3.74%. X8X emerges as the primary net recipient of volatility at −13.87%, followed by PaxGold at −4.00%.

For long-term volatility transmission (Table 4), HelloGold has a NET value of 1.80%, followed by BTC at 1.48%. X8X remains a net recipient with a NET value of −3.02%, and PaxGold has a NET value of −0.27%.

Overall, for short-term volatility, Bitcoin and HelloGold act as net transmitters, while X8X and PaxGold are net recipients. In the long-term horizon, HelloGold and BTC continue to be net transmitters of volatility, albeit to a lesser extent, while X8X remains the primary net recipient, and PaxGold exhibits a minimal net reception of volatility. These findings align with Huang et al. (2023) research, which similarly observed a time-frequency prolonged pattern of volatility transmission among Bitcoin and other financial markets. These outcomes suggest that factors such as the global pandemic and geopolitical tensions may have induced a fundamental shift in investor expectations, resulting in heightened long-term uncertainty and systemic risk.

3.2. Total dynamic volatility connectedness

For a better understanding of the influence of the COVID-19 health crisis on these interconnections across the network of markets, we lead a more dynamic analytical framework that considers the changing nature of the TACI over time (depicted with black color) and reflects how the roles of the studied markets within the network evolve illustrated in Figure 3.

This figure shows several notable features of the entire volatility transmission dynamics. It also demonstrates a breakdown into its short-term aspects (presented by the red-shaded area) and long-term components (depicted by the green-shaded area). The dominance of low-frequency response to shocks primarily drives the total volatility spillover, which aligns with the findings of Huang et al. (2023). In addition, we noticed a sharp decrease in TACI in late January 2020 and a considerable increase in March 2020, which coincided with the COVID-19 epidemic spreading rapidly worldwide, causing global financial market turmoil. Although COVID-19 continues to pose a threat, there has been a downward trend in total volatility transmission since its peak in April 2020. This may be due to the government's quick response to financial risks but a relatively inactive attitude toward controlling the pandemic. Finally, there has been a noticeable upward trend in the dynamic volatility transmission since February 2022, which may be related to market uncertainty caused by the Russia-Ukraine conflict.

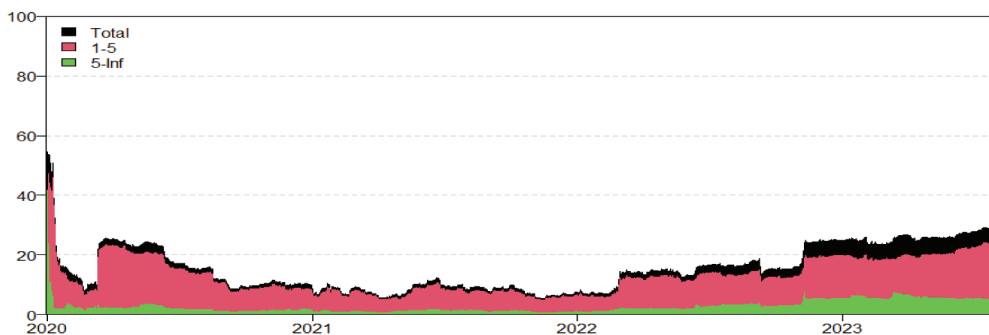


Figure 3. Total dynamic volatility connectedness.

3.3. Net total directional volatility connectedness

The net directional volatility connectedness in [Figure 4](#) exhibits time-dependent characteristics. Specifically, Bitcoin initially acted as a net receiver of volatility spillovers before January 2020 but transitioned into a net transmitter, particularly after February 2020. HelloGold and Bitcoin significantly increase their volatility spillover to other markets during this period. Conversely, the PaxGold market primarily functioned as a modest net transmitter before January 2020. After that, PaxGold and X8Xmarkets became prominent net receivers during the pandemic. Once more, the frequency decomposition results emphasize that short-term components play a pivotal role in directional volatility spillovers. This suggests that volatility transmission among cryptocurrency markets tends to occur over reduced time horizons.

This section delves into the time-frequency domain to examine the pairwise volatility spillovers between cryptocurrency markets, as depicted in [Figure 5](#). Focusing on the directional volatility spillovers involving Islamic Gold-backed cryptocurrencies and other financial assets, it is evident that Bitcoin predominantly transmits spillovers to X8X and PaxGold markets after January 2020, mainly in the short horizons. However, following the outbreak of COVID-19, HelloGold started transferring volatility to the Bitcoin market until December 2020 and reversed this pattern after January 2021. Additionally, HelloGold appears to significantly influence the volatility transmission to X8X and PaxGold for most of the observation period, especially in the short horizons. On the flip side, negligible contagion links are observed between the Islamic-backed-gold cryptocurrencies and Pax Gold. However, notable interaction is evident between Bitcoin and Islamic-backed-gold cryptocurrency markets. In addition, the net pairwise volatility interactions between the studied markets are relatively short-lived, reaffirming the conclusions drawn from our previous analyses.

The volatility-reducing effect of Islamic gold-backed cryptocurrency is incorporated by a comparative analysis of market reactions to traditional versus gold-backed Islamic cryptocurrencies during periods of financial stress. The empirical

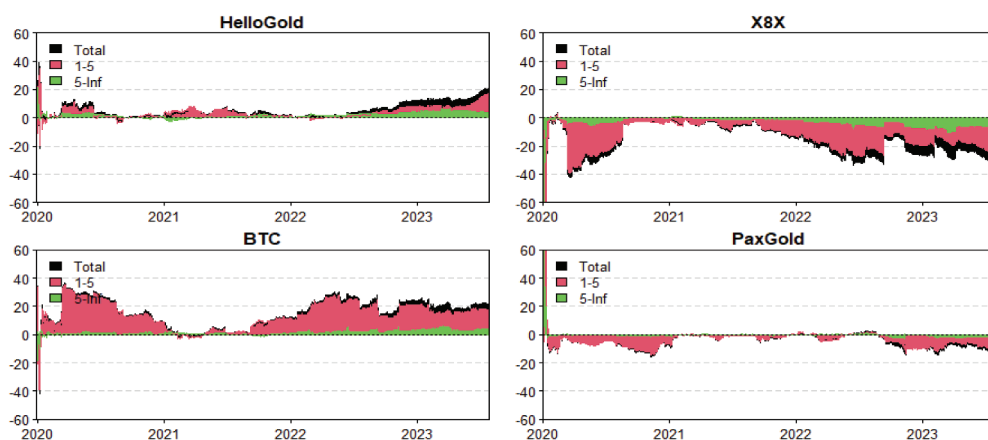


Figure 4. Dynamic total directional volatility connectedness.

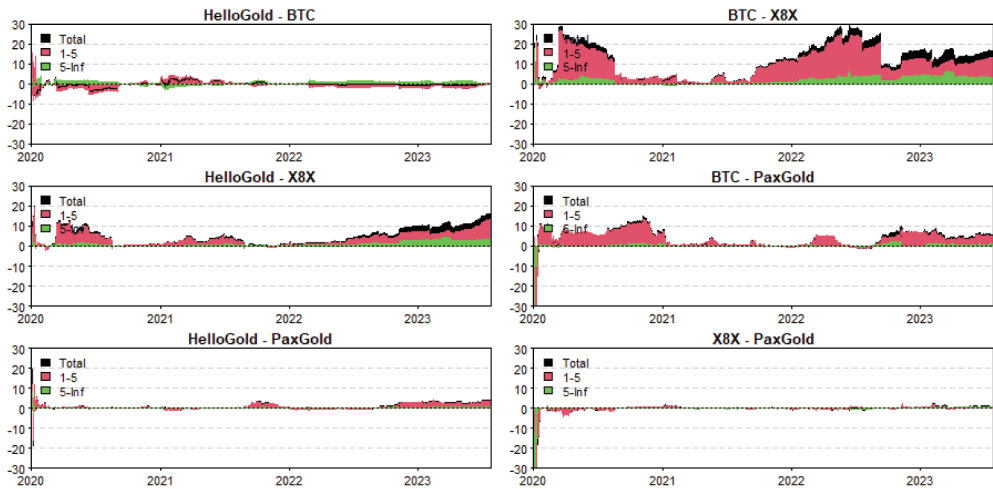


Figure 5. Dynamic net pairwise directional volatility connectedness.

findings indicate that markets incorporating Islamic gold-backed cryptocurrencies exhibit less volatility, supported by pairwise connectedness results in response to global financial shocks. These results are statistically significant and underscore the potential of Islamic financial instruments to contribute to more stable and resilient financial markets.

3.4. Robustness check

To validate the robustness of our research findings, we employed the Quantile Vector Autoregression (QVAR) model for robustness checks. We compared its outcomes with those from impulse response analysis at various time points in an earlier study. The QVAR model allows for analyzing the dynamic relationships between variables across different quantiles, providing a more comprehensive understanding of their interactions compared to the traditional VAR model. This approach is beneficial for capturing the behavior of variables under different market conditions or states of the economy. By examining the responses at various quantiles, we can assess the robustness of our results across a range of scenarios, ensuring the reliability of our findings. Thus, the QVAR model is well-suited for the robustness test in our study.

Figure 6 illustrates the responses of Islamic gold-backed cryptocurrencies to shocks from Bitcoin prices within the QVAR model framework. The figure reveals that the Bitcoin and HelloGold markets are net transmitters of shocks on X8X and PaxGold, especially at the beginning of the pandemic and the Russia-Ukraine crisis in the short term, with less pronounced effects in the long term. By comparing the results of pairwise volatility spillovers between each cryptocurrency market within the framework of the Quantile time-frequency connectedness approach, it is evident that the outcomes of the QVAR model are similar to those of the TVP-VAR model discussed earlier in the paper, confirming the robustness of our results.

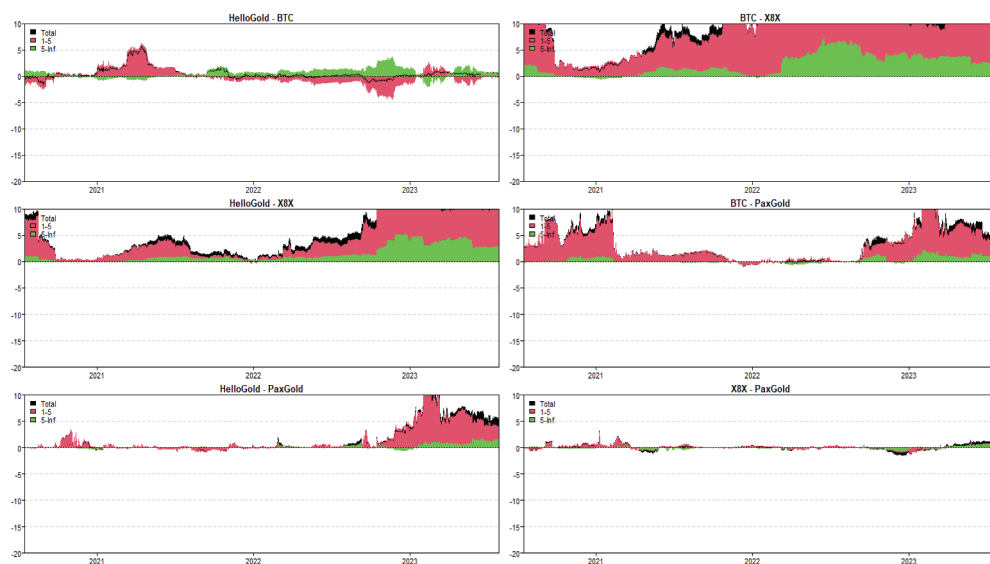


Figure 6. QVAR analysis for results robustness.

4. Discussion

This paper utilizes a TVP-VAR frequency connectedness model to examine the volatility transmission between Islamic and conventional cryptocurrency markets during the global pandemic, providing a comprehensive understanding of the topic. The network volatility transmission is a significant source of average total spillover among cryptocurrency markets, indicating a more integrated global financial system and increasing systemic risk contagion. The total volatility transmission pattern is time-varying, with a remarkable peak during the outbreak of COVID-19 and an increased level after the Russia-Ukraine conflict. The results also illustrate that the dominance of low-frequency response to shocks primarily drives the total volatility spillover. In addition, Bitcoin and HelloGold act as net transmitters of the volatility transmission process depending on different periods, particularly during the global pandemic, implying that the Islamic cryptocurrency market is becoming more integrated with conventional ones. These research findings carry profound implications for the role of Islamic cryptocurrencies in advancing sustainability. By shedding light on the intricate dynamics of volatility spillovers involving Islamic Gold-backed cryptocurrencies, such as XBX and HelloGold, and the cryptocurrency giant Bitcoin, we unveil a critical aspect of the global financial landscape. As we navigate an era characterized by economic uncertainties, the insights derived from this study underscore the potential of Islamic cryptocurrencies in fostering financial stability and aligning with SDGs. With their unique features and abilities to bridge traditional finance with the emerging digital economy, these digital assets could contribute significantly to achieving Islamic principles and promoting financial stability.

The dynamic transmission of volatility across Islamic and conventional gold-backed cryptocurrencies, as revealed through our application of the Time-Varying Parameter Vector Autoregression (TVP-VAR) frequency connectedness approach, offers profound insights into the stability and resilience of financial markets. The roles of Bitcoin and

HelloGold as net transmitters and X8X and PaxGold as net recipients of volatility underscore the interconnectedness of digital asset markets and their influence on financial stability.

Volatility, often viewed through a narrow lens as a measure of risk, holds broader implications for sustainable finance. High volatility can signal uncertainty and instability, potentially deterring investment in sustainable projects that require stable, predictable returns. Conversely, understanding the sources and receivers of volatility within the financial system can inform strategies to mitigate risk, promote resilience, and support the financing of sustainable development initiatives.

Our findings illuminate the potential of Islamic gold-backed cryptocurrencies to contribute to a more sustainable and resilient financial system. By acting as both transmitters and recipients of volatility, these assets reveal the complexity of their roles within the financial ecosystem.

The study's insights into the volatility transmission among these cryptocurrencies during a period marked by significant global upheavals, including the COVID-19 pandemic, highlight the need for financial systems that can withstand external shocks. Islamic gold-backed cryptocurrencies, with their inherent stability linked to gold and adherence to ethical financial practices, offer a model for such resilience. This model is particularly relevant to sustainable finance, prioritizing long-term stability and the capacity to support projects that advance the SDGs. In light of our analysis, future research should delve further into how the stabilizing features of Islamic gold-backed cryptocurrencies can be leveraged within the broader framework of sustainable finance. Specifically, empirical studies could explore how these assets affect the overall resilience of financial portfolios during periods of economic stress, contributing to a more stable investment climate conducive to sustainable development.

For practitioners, integrating Islamic gold-backed cryptocurrencies into investment strategies presents an opportunity to align with sustainable finance goals. Financial institutions and policymakers can consider these assets part of a diversified portfolio that seeks to minimize risk and contribute to a more equitable and sustainable world.

5. Conclusion and recommendations

5.1. Conclusion

This study explores the volatility transmission between Islamic and conventional cryptocurrency markets using a TVP-VAR frequency connectedness model, highlighting the interconnectedness of these markets and their impact on financial stability. The research shows that Bitcoin and HelloGold act as net transmitters of volatility, particularly during global crises, suggesting that the Islamic cryptocurrency market is increasingly integrated with conventional ones. This integration carries implications for the role of Islamic cryptocurrencies in promoting sustainability. The study sheds light on the dynamics of volatility spillovers involving Islamic Gold-backed cryptocurrencies, such as X8X and HelloGold, and Bitcoin, revealing a critical aspect of the global financial landscape. The insights from this study underscore the potential of Islamic cryptocurrencies in fostering financial stability and aligning with Sustainable Development Goals (SDGs). The dynamic transmission of volatility across Islamic and conventional gold-backed cryptocurrencies offers profound insights into the stability and resilience of financial markets. Islamic gold-backed cryptocurrencies not only provide a compliant alternative for investors seeking

stability in their portfolios but also promote broader financial inclusion and sustainability, which are critical components of the SDGs.

Future research should delve further into how the stabilizing features of Islamic gold-backed cryptocurrencies can be leveraged within the broader framework of sustainable finance. Specifically, empirical studies could explore how these assets affect the overall resilience of financial portfolios during periods of economic stress, contributing to a more stable investment climate conducive to sustainable development.

5.2. Implications and recommendations

For investors and portfolio managers, the importance of diversification and advanced risk management strategies is paramount. The interconnectedness of cryptocurrency markets with traditional financial systems necessitates a cautious approach to asset allocation, particularly considering the hedging potential of Islamic gold-backed cryptocurrencies against market volatility. These assets should be considered part of a broader strategy to enhance portfolio resilience, focusing on ethical and sustainable investment options. Nevertheless, the academic and research community is encouraged to delve deeper into the temporal variations in risk propagation among cryptocurrencies, aiming to develop financial models that accurately reflect these dynamics. Longitudinal studies assessing the impact of macroeconomic factors on cryptocurrency markets could provide invaluable insights, contributing to more sophisticated risk management and investment strategies. In addition, financial institutions and sustainability advocates are positioned to leverage these insights to promote sustainable finance. By increasing awareness and understanding of Islamic gold-backed cryptocurrencies and their alignment with ethical investing principles, these stakeholders can support the integration of cryptocurrencies into sustainable investment portfolios, advancing broader societal and environmental objectives. By aligning strategies with sustainability goals and enhancing financial stability, the financial ecosystem can contribute to a more sustainable and equitable global financial system, leveraging the unique potential of cryptocurrencies to support economic stability, financial inclusion, and climate action. Further research could explore the impact of regulatory changes on the volatility transmission among Islamic and conventional cryptocurrencies, examining how different regulatory environments influence market dynamics and the integration of these assets into the financial system. Besides, there is a need for longitudinal studies that assess the long-term implications of integrating Islamic cryptocurrencies into sustainable investment strategies, examining their impact on financial stability, ethical investing, and the achievement of Sustainable Development Goals. Policymakers in China and other developing countries are encouraged to explore regulatory frameworks that facilitate the integration of Islamic gold-backed cryptocurrencies into their financial systems. By fostering an environment that supports ethical and sustainable finance, these countries can enhance their attractiveness to global investors who are increasingly focused on responsible investment practices. Policymakers and regulators are advised to align financial regulations with Sustainable Development Goals (SDGs), fostering practices that contribute to sustainable finance.

With a rapidly growing interest in digital finance and sustainable development for Chinese and developed countries, the insights from this study on the stabilizing potential of Islamic gold-backed cryptocurrencies can inform strategies to build resilient financial

systems that support sustainable development. These countries can leverage the findings to explore the integration of ethical and stable digital assets into their financial markets, contributing to economic stability and sustainable growth.

Acknowledgement

The authors extend their appreciation to the Deanship of Research and Graduate Studies at King Khalid University for funding this work through small group research under grant number RGP1/162/45.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Dr. Emna Mnif actively participates in pioneering research as a member of LATIGE-Lab., a research organization dedicated to Technology, Governance, and Entrepreneurship. Her research interests span a broad spectrum of finance and data science topics, highlighting her interdisciplinary approach. Dr. Mnif has made noteworthy contributions to the academic community by publishing numerous scholarly papers in respected international refereed journals.

Dr. Yomna Daoud is a Doctor of Finance from the University of Sfax, Tunisia. Dr. Daoud is an active participant in pioneering research as a member of LEG-Lab, a research laboratory in economics and management. Her research interests cover various financial topics, with special focus on business research, Islamic finance, and financial development. Dr. Daoud has made significant academic contributions through publications in prestigious international peer-reviewed journals.

Akrout Zied is Professor in Economics Sciences, Economics department, College of Business, King Khalid University, Abha, Kingdom of Saudi Arabia. He has carried out research and consulting activities in the field of Applied Micro econometrics, Development Economics, Labour Economics and Tourism Economics.

Anis Jarboui is a Professor in Business Administration at the University of Sfax. He holds a Ph.D. in Finance from the University of Nice Sophia Antipolis- France (Université Côte d'Azur 2004) and a HDR degree (Habilitation for Supervising Doctoral Research) in 2008. He was Dean (College of Business Anis Administration) of the Higher Institute of Business Administration of Sfax from 2011 to 2017. Pr. Jarboui has recently published many academic papers (more than 100 papers) in international refereed journals.

ORCID

Yomna Daoud  <http://orcid.org/0000-0002-8809-4923>

Anis Jarboui  <http://orcid.org/0000-0002-4811-6729>

Ethical approval

The authors declare that the experiments comply with the current laws of the country in which they were performed.

References

- Abakah, E. J. A., A. K. Tiwari, S. Ghosh, and B. Doğan. 2023. "Dynamic Effect of Bitcoin, Fintech and Artificial Intelligence Stocks on Eco-Friendly Assets, Islamic Stocks and Conventional Financial Markets: Another Look Using Quantile-Based Approaches." *Technological Forecasting & Social Change* 192:122566. <https://doi.org/10.1016/j.techfore.2023.122566>.
- Ali, F., E. Bouri, N. Naifar, S. J. H. Shahzad, and M. AlAhmad. 2022. "An Examination of Whether Gold-Backed Islamic Cryptocurrencies are Safe Havens for International Islamic Equity Markets." *Research in International Business and Finance* 63:101768. <https://doi.org/10.1016/J.RIBAF.2022.101768>.
- Aloui, C., H. ben Hamida, and L. Yarovaya. 2021. "Are Islamic Gold-Backed Cryptocurrencies Different?" *Finance Research Letters* 39:101615. <https://doi.org/10.1016/j.frl.2020.101615>.
- Aloui, C., H. B. Hamida, and L. Yarovaya. 2021. "Are Islamic Gold-Backed Cryptocurrencies Different?" *Finance Research Letters* 39 (March) 101615. <https://doi.org/10.1016/j.frl.2020.101615>.
- Antonakakis, N., I. Chatziantoniou, and D. Gabauer. 2020. "Refined Measures of Dynamic Connectedness Based on Time-Varying Parameter Vector Autoregressions." *Journal of Risk and Financial Management* 13 (4): 84. <https://doi.org/10.3390/jrfm13040084>.
- Baruník, J., and T. Křehlík. 2018. "Measuring the Frequency Dynamics of Financial Connectedness and Systemic Risk*." *Journal of Financial Econometrics* 16 (2): 271–296. <https://doi.org/10.1093/jjfne/nby001>.
- Chatziantoniou, I., D. Gabauer, and R. Gupta. 2021. "Integration and Risk Transmission in the Market for Crude Oil: A Time-Varying Parameter Frequency Connectedness Approach." University of Pretoria Department of Economics Working Paper Series, June. https://www.up.ac.za/media/shared/61/WP/wp_2021_47.zp209709.pdf.
- Diebold, F. X., and K. Yilmaz. 2012. "Better to Give Than to Receive: Predictive Directional Measurement of Volatility Spillovers." *International Journal of Forecasting* 28 (1): 57–66. <https://doi.org/10.1016/J.IJFORECAST.2011.02.006>.
- Diebold, F. X., and K. Yilmaz. 2014. "On the Network Topology of Variance Decompositions: Measuring the Connectedness of Financial Firms." *Journal of Econometrics* 182 (1): 119–134. <https://doi.org/10.1016/J.JECONOM.2014.04.012>.
- Elliott, G., T. J. Rothenberg, and J. H. Stock. 1996. "Efficient Tests for an Autoregressive Unit Root." *Econometrica* 64: 813–836.
- Elsayed, A. H., and M. H. Helmi. 2021. "Volatility Transmission and Spillover Dynamics Across Financial Markets: The Role of Geopolitical Risk." *Annals of Operations Research* 305 (1): 1–22. <https://doi.org/10.1007/s10479-021-04081-5>.
- García-Monleón, F., I. Danvila-Del-Valle, and F. J. Lara. 2021. "Intrinsic Value in Crypto Currencies." *Technological Forecasting & Social Change* 162 (October 2020): 120393. <https://doi.org/10.1016/j.techfore.2020.120393>.
- Grover, P., A. K. Kar, and P. V. Ilavarasan. 2019. "Impact of Corporate Social Responsibility on Reputation—Insights from Tweets on Sustainable Development Goals by CEOs." *International Journal of Information Management* 48:39–52. <https://doi.org/10.1016/j.ijinfomgt.2019.01.009>.
- Hassan, M. K., H. G. Djajadikerta, T. Choudhury, and M. Kamran. 2021. "Safe Havens in Islamic Financial Markets: COVID-19 versus GFC." *Global Finance Journal* 54:100643. <https://doi.org/10.1016/J.GFJ.2021.100643>.
- Huang, J., B. Chen, Y. Xu, and X. Xia. 2023. "Time-Frequency Volatility Transmission Among Energy Commodities and Financial Markets During the COVID-19 Pandemic: A Novel TVP-VAR Frequency Connectedness Approach." *Finance Research Letters* 53:103634. <https://doi.org/10.1016/J.FRL.2023.103634>.
- Jarque, C. M., and A. K. Bera. 1980. "Efficient Tests for Normality, Homoscedasticity and Serial Independence of Regression Residuals." *Economics Letters* 6 (3): 255–259. [https://doi.org/10.1016/0165-1765\(80\)90024-5](https://doi.org/10.1016/0165-1765(80)90024-5).
- Jiang, X. 2020. "Digital Economy in the Post-Pandemic Era." *Journal of Chinese Economic & Business Studies* 18 (4): 333–339. <https://doi.org/10.1080/14765284.2020.1855066>.

- Koeswandana, N. A., and F. A. Sugino. 2023. "Intention to Use Cryptocurrency: Social and Religious Perspective." *Jurnal Ekonomi & Keuangan Islam* 9 (1): 91–103. <https://doi.org/10.20885/JEKI.vol9.iss1.art7>.
- Lahmiri, S., and S. Bekiros. 2019. "Decomposing the Persistence Structure of Islamic and Green Crypto-Currencies with Nonlinear Stepwise Filtering." *Chaos, Solitons & Fractals* 127:334–341. <https://doi.org/10.1016/j.chaos.2019.07.012>.
- Lu, R., W. Xu, H. Zeng, and X. Zhou. 2023. "Volatility Connectedness Among the Indian Equity and Major Commodity Markets Under the COVID-19 Scenario." *Economic Analysis & Policy* 78:1465–1481. <https://doi.org/10.1016/j.eap.2023.05.020>.
- Maghyereh, A., B. Awartani, and A. Hassan. 2018. "Can Gold Be Used as a Hedge Against the Risks of Sharia-Compliant Securities? Application for Islamic Portfolio Management." *Journal of Asset Management* 19 (6): 394–412. <https://doi.org/10.1057/s41260-018-0090-y>.
- Mnif, E., K. Mouakhar, and A. Jarboui. 2022. "The Co-Movements of Faith-Based Cryptocurrencies in Periods of Pandemics." *Review of Financial Economics* 40 (3): 300–311. <https://doi.org/10.1002/rfe.1154>.
- Rabbani, M. R., A. Bashar, N. Nawaz, S. Karim, M. A. M. Ali, H. U. Rahiman, and M. S. Alam. 2021. "Exploring the Role of Islamic Fintech in Combating the Aftershocks of Covid-19: The Open Social Innovation of the Islamic Financial System." *Journal of Open Innovation* 7 (2): 136. <https://doi.org/10.3390/joitmc7020136>.
- Said, M. S., H. A. Annuar, and H. B. Hamdan. 2019. "An Investigation into the Financial Sustainability of Islamic Saving, Credit Cooperative Society (SACCOS) in Tanzania." *International Journal of Ethics and Systems* 35 (2): 242–259. <https://doi.org/10.1108/IJOES-11-2018-0159>.
- Shaikh, I. M., M. A. Qureshi, K. Noordin, J. M. Shaikh, A. Khan, and M. S. Shahbaz. 2020. "Acceptance of Islamic Financial Technology (FinTech) Banking Services by Malaysian Users: An Extension of Technology Acceptance Model." *Foresight* 22 (3): 367–383. <https://doi.org/10.1108/FS-12-2019-0105>.
- Siswanto, D., R. Handika, and A. F. Mita. 2020. "The Requirements of Cryptocurrency for Money, an Islamic View." *Heliyon* 6 (1): e03235. <https://doi.org/10.1016/j.heliyon.2020.e03235>.
- Yang, Q., Q. Du, A. Razzaq, and Y. Shang. 2022. "How Volatility in Green Financing, Clean Energy, and Green Economic Practices Derive Sustainable Performance Through ESG Indicators? A Sectoral Study of G7 Countries." *Resources Policy* 75:102526. <https://doi.org/10.1016/j.resourpol.2021.102526>.
- Yousaf, I., and L. Yarovaya. 2022. "Spillovers Between the Islamic Gold-Backed Cryptocurrencies and Equity Markets During the COVID-19: A Sectorial Analysis." *Pacific-Basin Finance Journal* 71:101705. <https://doi.org/10.1016/J.PACFIN.2021.101705>.